STATISTICAL SCRAPING AT STATISTICS HESSE: PAST, PRESENT, FUTURE

Tobias Gramlich, Hessisches Statistisches Landesamt

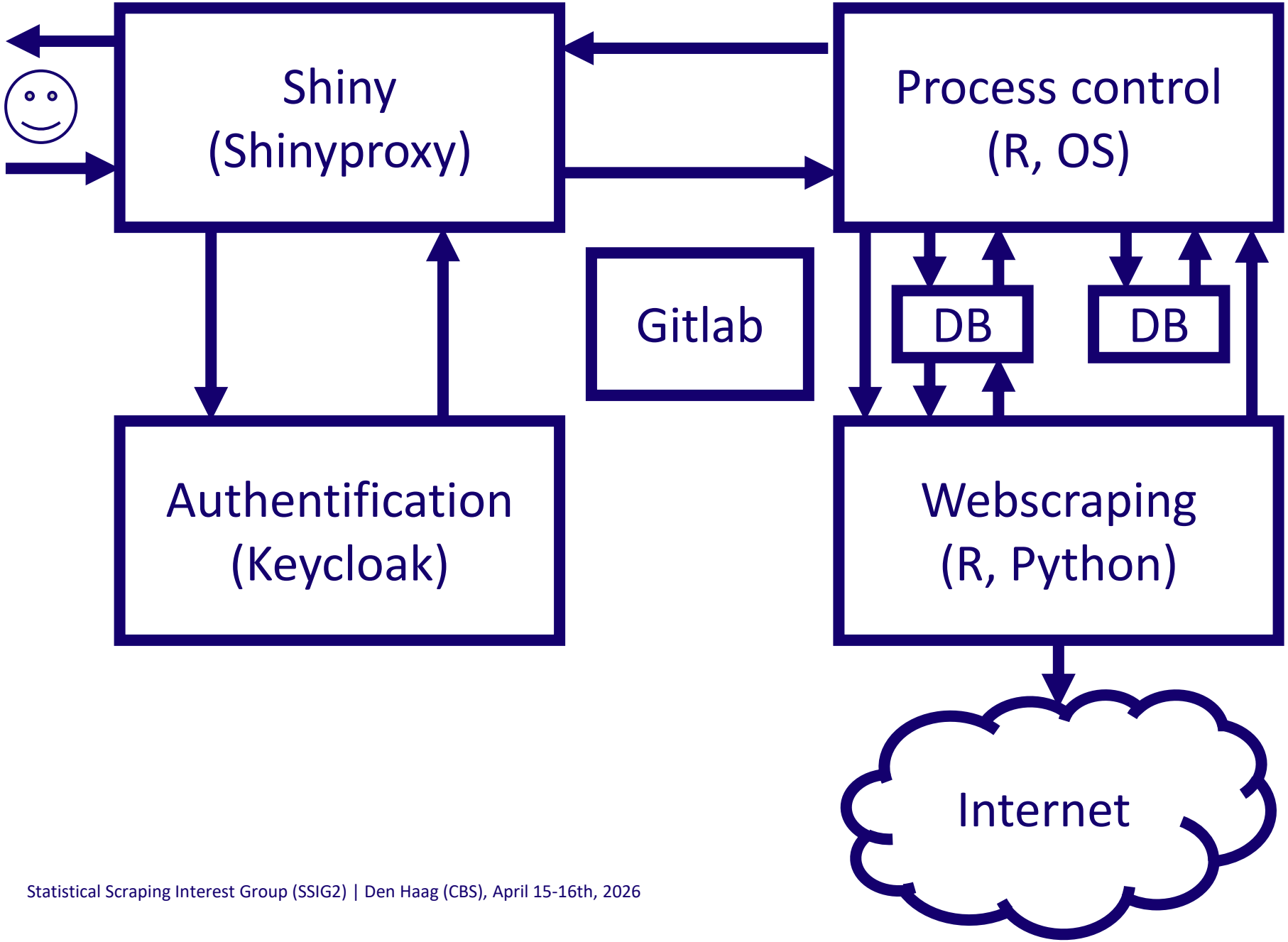
Statistical Scraping Interest Group (SSIG2) | Den Haag (CBS), April 15-16th, 2026

01
PAST

PAST

Development of a system for finding URLs

- Started in 2018 (!)
- Using Google Custom Search API
- Participation in several related ESSnet projects
  - Big Data I, Big Data II, Web Intelligence Network WIN
  - MNO data, „webscraping“ (URL finding; hotels, newly constructed houses)
  - A lot of input from colleagues from other statistical offices (CBS, Statistik Austria, ISTAT, GUS, …)
  - Hopefully input for others, too



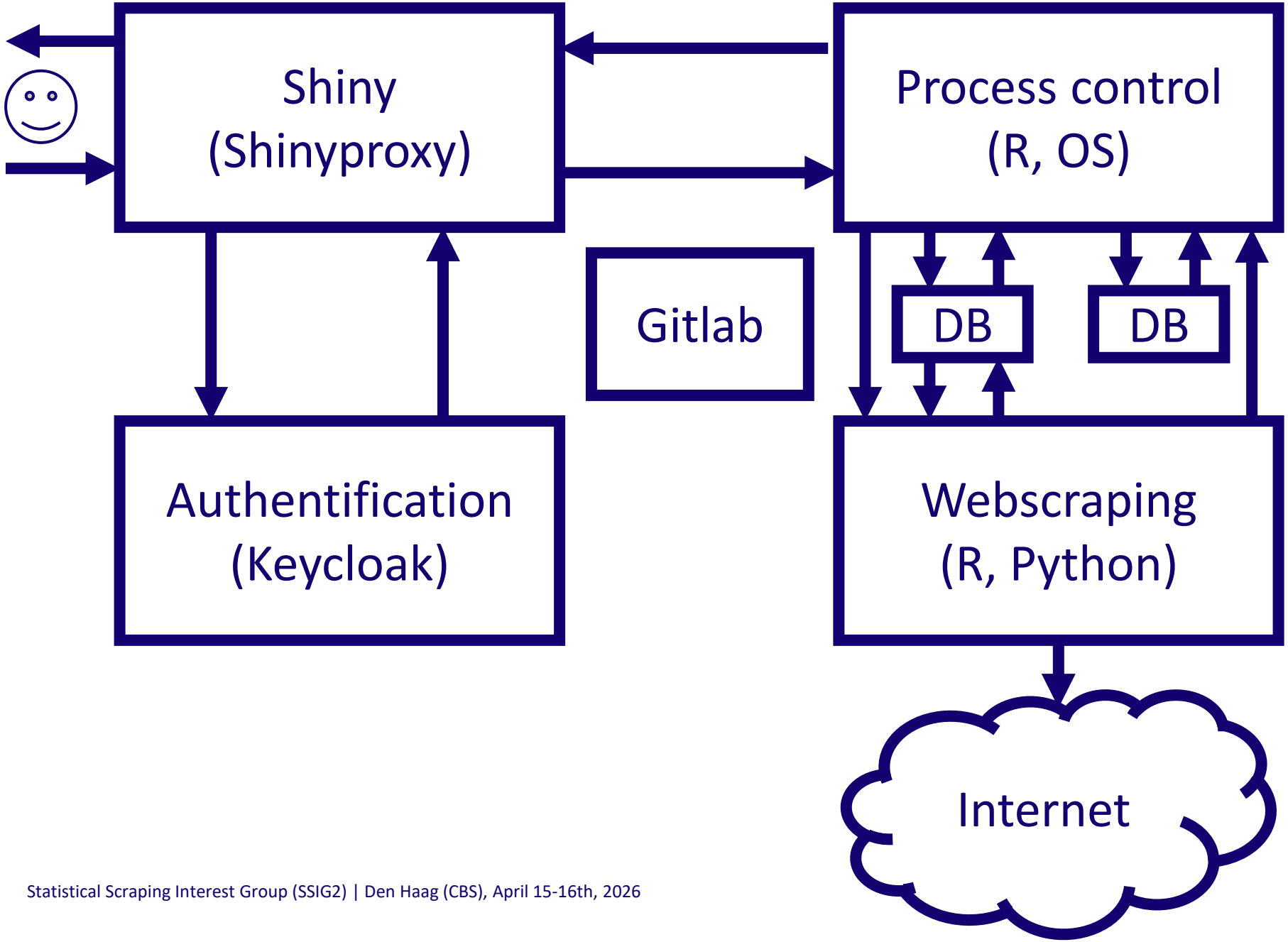
Synchronization of files

Status files:
StatusShiny
StatusIntern
StatusWebscraping

Input files:
SRB (sample), keywords

Output files:
URLs, keyword matches

02 PRESENT



Synchronization of files

Status files:
StatusShiny
StatusIntern
StatusWebscraping

Input files:
SRB (sample), keywords

Output files:
URLs, keyword matches

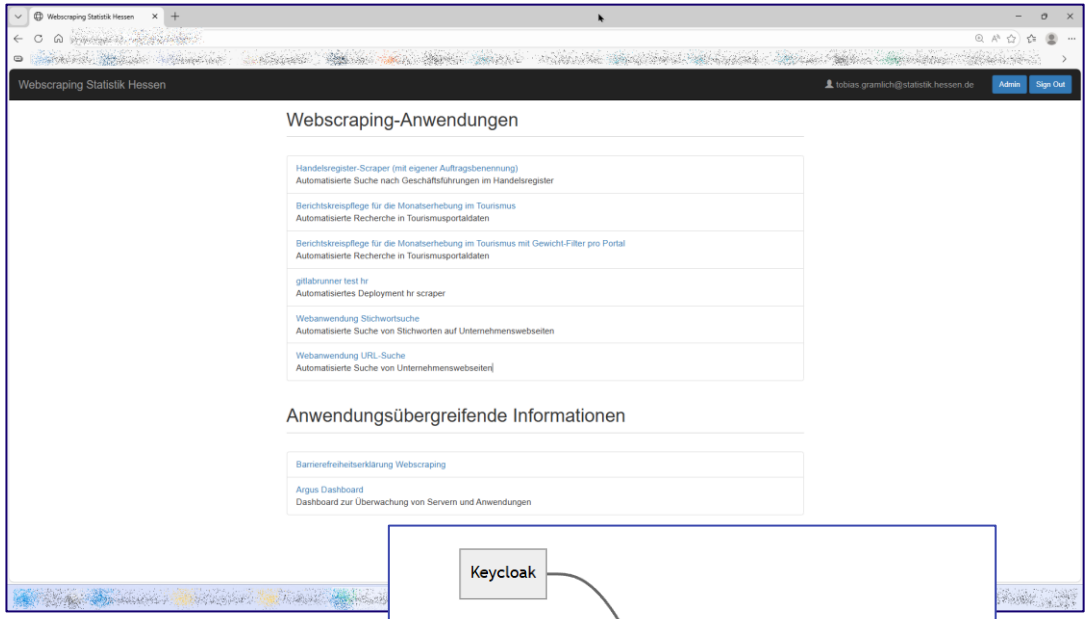
PRESENT

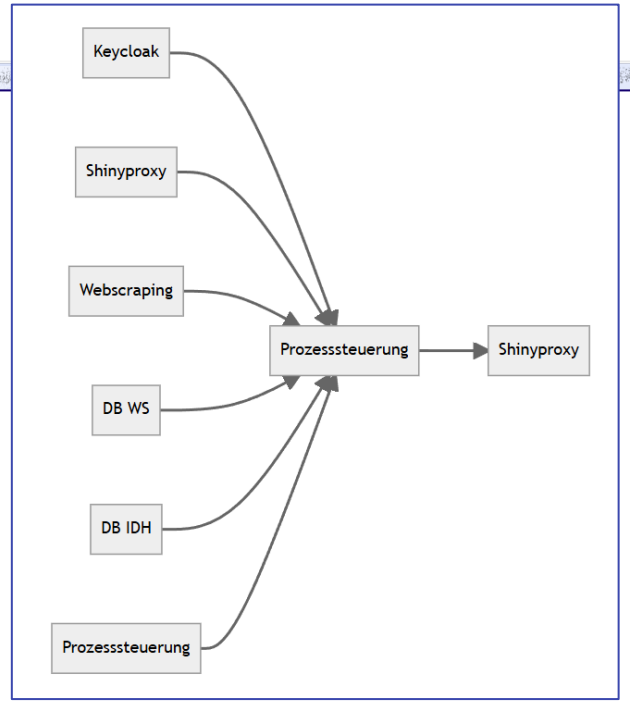
Things to think about and to do at the moment

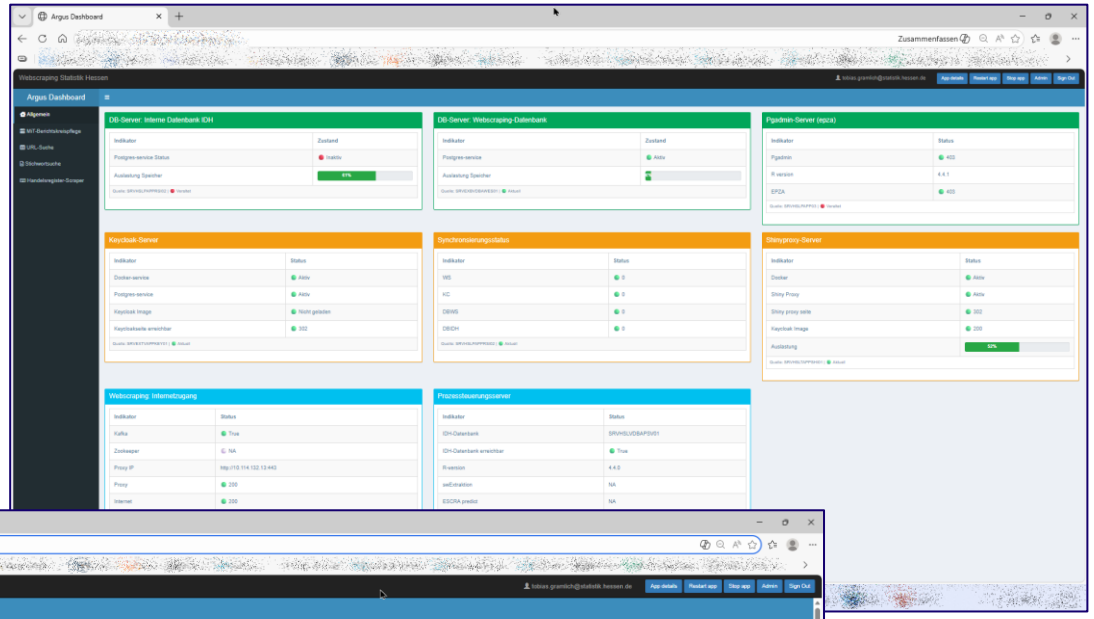
Since June 2025, run and maintain a production system for searching

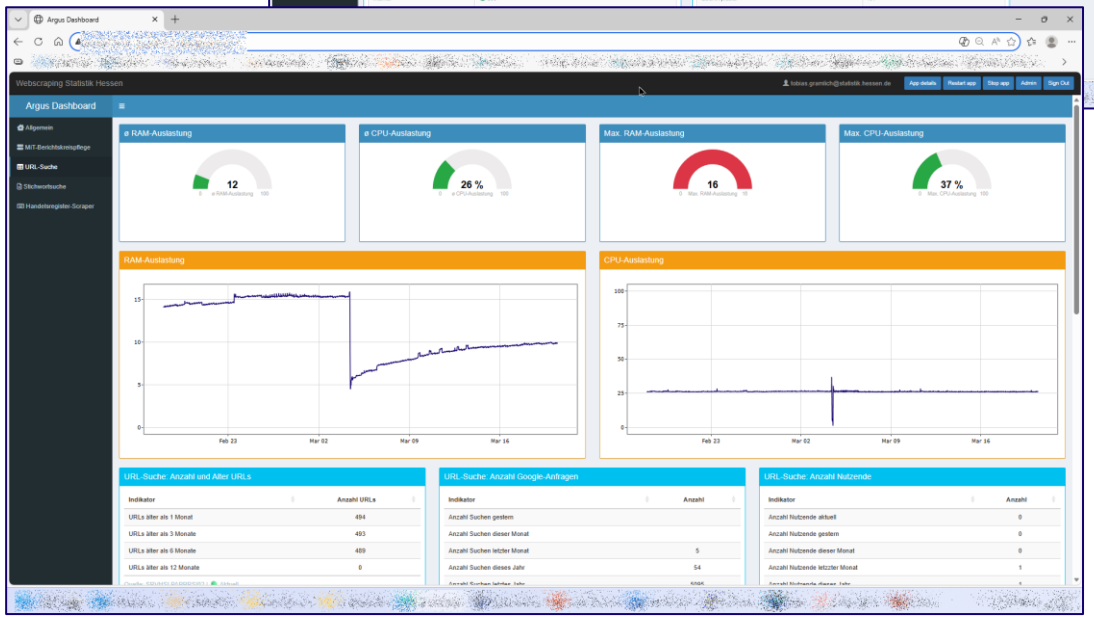
- URLs (domains) to units from the SBR
- for keywords (not) appearing on scraped webpages
- (automatically) for information from Commercial Trade Registry documents: Name(s) of managing director(s) for legal documents

- System administration, i.e.:
  - Updating OS and application software
  - Managing users and system ressources
  - Fixing bugs
- Active monitoring of system


Webscraping Statistik Hessen
tobias.gramlich@statistik.hessen.de
Admin
Sign Out
Webscraping-Anwendungen
Handelsregister-Scraper (mit eigener Auftragsbenennung)
Automatisierte Suche nach Geschäftsführungen im Handelsregister
Berichtskreispflege für die Monatserhebung im Tourismus
Automatisierte Recherche in Tourismusportaldaten
Berichtskreispflege für die Monatserhebung im Tourismus mit Gewicht-Filter pro Portal
Automatisierte Recherche in Tourismusportaldaten
gitlabrunner test hr
Automatisiertes Deployment hr scraper
Webanwendung Stichwortsuche
Automatisierte Suche von Stichworten auf Unternehmenswebseiten
Webanwendung URL-Suche
Automatisierte Suche von Unternehmenswebseiten
Anwendungsübergreifende Informationen
Barrierefreiheitserklärung Webscraping
Argus Dashboard
Dashboard zur Überwachung von Servern und Anwendungen


Keycloak
Shinyproxy
Webscraping
DB WS
DB IDH
Prozesssteuerung
Prozesssteuerung
Shinyproxy


Argus Dashboard
Webscraping Statistik Hessen
DB-Server: Interne Datenbank IDH
DB-Server: Webscraping-Datenbank
Pgadmin-Server (epca)
Keycloak-Server
Synchronisierungsstatus
Shinyproxy-Server
Webscraping: Internetzugang
Prozesssteuerungsserver


Argus Dashboard
Webscraping Statistik Hessen
ø RAM-Auslastung
12
ø CPU-Auslastung
26 %
Max. RAM-Auslastung
16
Max. CPU-Auslastung
37 %
RAM-Auslastung
CPU-Auslastung
URL-Suche: Anzahl und Alter URLs
URL-Suche: Anzahl Google-Anfragen
URL-Suche: Anzahl Nutzende

NUMBERS

More or less concrete plans and goals

Some numbers from the monitoring

	zeitstempel text	suchen1 bigint	suchen2 bigint	suchen3 bigint	suchen4 bigint	suchen5 bigint
1	2026-04-14 21:04	13764	47567	180121	515169	819042

	nutzende bigint	nutzende1 bigint	nutzende2 bigint	nutzende3 bigint	nutzende4 bigint	nutzende5 bigint	zeitstempel text
1	1	2	2	4	7	17	2026-04-14 21:04

	urls bigint	gestern bigint	alter1 bigint	alter2 bigint	alter3 bigint	alter4 bigint	alter5 bigint	alter6 bigint	alter7 bigint	zeitstempel text
1	113983	7368	20244	92366	104535	110054	113983	3929	0	2026-04-14 21:04

	zeitstempel text	laufend bigint	abgeschlossen1 bigint	abgeschlossen2 bigint	abgeschlossen3 bigint	abgeschlossen4 bigint	abgeschlossen5 bigint	abgebrochen1 bigint	abgebrochen2 bigint	abgebrochen3 bigint	abgebrochen4 bigint	abgebrochen5 bigint	fehler1 bigint	fehler2 bigint	fehler3 bigint	fehler4 bigint	fehler5 bigint
1	2026-04-14 21:...	2	2	19	63	194	743	0	0	0	0	64	0	0	1	2	4

03
FUTURE

FUTURE

More or less concrete plans and goals

Where to go next?

- Uploading „known to be correct“ URLs to DB by user
  - „update“ existing URLs in DB?
- Updating (verifying) URLs in DB
  - Automatically, e.g. after 12 months
  - Only re-checking HTTP request status for URL from DB, or scraping and matching URL to SBR data again?
- Extract imprint information from webpages
  - Trade registry does not contain all types of businesses
  - In principle, there is a legal obligation of providing imprint on website (§5 Telemedia Act)

04
OWS/OWI

OWI/OWS

More or less concrete plans and goals

Use OWS/OWI

- Instead of Google Custom Search API for URL-finding?
  - Overcome daily limit of 10.000 searches
  - Overcome dependency on commercial search engine provider from outside EU
- For updating / verifying URLs in DB?

- Technical requirements?
- Comparability of results? Differences to expect?

THANKS

Contact

Tobias Gramlich
Hessisches Statistisches Landesamt
tobias.gramlich@statistik.hessen.de
Tel. +49 (0) 611 3802-845

HESSEN